BTS –V(15) –11.17–1538 Reg. No. [ ]

C

B.Tech. Degree V Semester Examination November 2017

ME 15-1504 THERMAL ENGINEERING

(2015 Scheme)

Time : 3 Hours Maximum Marks : 60

PART A

(Answer *ALL* questions)

$(10 \times 2 = 20)$

I. (a) What do you mean by stoichiometric air-fuel ratio?

(b) Explain higher heating value (HHV) and lower heating value (LHV) of fuel.

(c) Draw and explain the valve timing diagram of a four stroke SI engine.

(d) What is a heat balance sheet?

(e) Draw and explain the actual cycle of a four stroke CI engine on p-v diagram.

(f) Differentiate between super charging and turbo charging of IC engines.

(g) Discuss the characteristics of stratified charged engine.

(h) What are the different methods of fuel rating?

(i) What is ignition lag? State the factors influencing it.

(j) What does HUCR stand for? Explain.

PART B

$(4 \times 10 = 40)$

II. (a) The percentage composition of sample of liquid fuel by weight is, C=84.8%, H_2=15.2%. Calculate (i) The weight of air needed for the combustion of 1 kg of fuel. (ii) The volumetric composition of the products of combustion if 15% excess air is supplied. (5)

(b) Liquid C_8H_{18} is completely burned in a continuous flow system with 320% theoretical air. Fuel is supplied at 25°C and air enters at 500 K. Estimate the adiabatic flame temperature. (5)

OR

III. (a) Determine the gravimetric analysis of the product of complete combustion of acetylene (C_2H_2) with 125% stoichiometric air. (5)

(b) The dry flue gas analysis by volume was obtained as CO_2=13%, CO = 0.3%, O_2=6%, N_2=80.7%. The coal analysis by weight was reported as C = 62.4%. H_2=4.2%, O_2=4.5%, moisture=15% and ash=13.9%. Estimate (i) Theoretical air required to burn 1 kg of coal (ii) Weight of air actually supplied per kg of coal (iii) The amount of excess air supplied per kg of coal burnt. (5)

IV. (a) A 4 stroke cylinder engine having bore = 80 mm and stroke = 100 mm was tested at 3500 rpm and following data were recorded. Fuel consumption = 300 gm/minute, indicated mean effective pressure = 1 MPa, engine torque developed = 140 Nm and the calorific value of fuel = 42000 kJ/kg, find Indicate Power mechanical efficiency and brake thermal efficiency of the engine. (5)

(b) A four cylinder engine has a total swept volume of 1.084 litres and develops a Brake Power of 23.4 kW at 3200 rpm. Calculate the brake mean effective pressure. (5)

OR

V. (a) While performing Morse test on a 4 cylinder, 4 stroke petrol engine, the following results were obtained at a particular throttle setting and speed. BP with all cylinders working = 32.2 kW, BP with cylinder no.1 cut out = 22 kW, BP with cylinder no.2 cut out = 21.8 kW, BP with cylinder no.3 cut out = 22.2 kW, BP with cylinder no.4 cut out = 22.8 kW. Determine IP and mechanical efficiency of the engine. (6)

(b) What would be the percentage change in the efficiency of an Otto cycle having a compression ratio of 10, when the specific heat C_v decreased by 1.5%. (4)

VI. (a) Name different methods of engine cooling. Explain the pump circulation cooling system and compare the same with the thermosyphon system. (5)

(b) What are the objectives of lubricating an engine? Briefly explain the different methods of engine lubrication. (5)

OR

VII. (a) Why is a governor absolutely necessary in case of diesel engine unlike petrol engine? Explain the working principle of various types of governors. (5)

(b) Draw and explain the components and working of a battery ignition system used for a 4 cylinder SI engine. (5)

VIII. (a) State various conditions affecting combustion chamber design and explain the same in detail. (5)

(b) Describe briefly the various theories of detonation. Give detailed account of the factors that influence detonation in IC engines. (5)

OR

IX. (a) Explain the term diesel knock. Describe how it is produced and compare the same with detonation in petrol engine. (5)

(b) Explain with figure the various types of combustion chambers for diesel engines. (5)

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